

Airbag ECU-Functional Safety Project

Project Planning Report | Rafsun (u41213), Project Manager | Hochschule Harz | Summer 2026

1. Project Overview

In compliance with ISO 26262 Functional Safety (FuSa) guidelines, this project creates an Arduino-based software-based Airbag Electronic Control Unit (ECU). Hazard analysis, ASIL determination, system architecture, Arduino software implementation, hardware concept design, testing and validation, and documentation management are all included in the scope. Hardware is seen as a defined interface assumption; physical hardware assembly is outside the preview.

2. Team Structure & Responsibilities

S/L	Role	Member(s)	Key Responsibility
1	Project Manager	Rafsun (u41213)	Planning, coordination, final submission
2	Safety Analysis Team (TUV Safety Authority)	Shakthi Prasad Gangi Reddy (u39436) Nirmal Patil (u39349)	Hazard analysis, ASIL determination
3	System Architecture Team	Huy Nguyen (u40313) Rafii Prasetya (u40280)	System design, block diagrams
4	Software Team	Atashi (u39941)	Arduino coding, logic implementation
5	Hardware Concept Engineer	Tanmay Aher (u39421)	Hardware design concept (no physical build)
6	Testing & Validation	Partha Pratim Sutradhar (u39352) Ashrafull Islam (u39428)	Test planning, execution, reporting
7	Documentation & Git Manager	Shruti (u39431)	Documentation, file organization, Git

3. Project Schedule

The project begins on April 20, 2026, and lasts for seven weeks. The order of tasks ensures that upstream deliverables (hazard analysis, architecture) are finished before downstream work (software, testing) starts.

Role	Start (Week)	Duration	Active Weeks
Project Manager	Week 1 (20 Apr)	7 weeks	W1 - W7
Safety Analysis Team	Week 1 (20 Apr)	3 weeks	W1 - W3
System Architecture Team	Week 1 (20 Apr)	3 weeks	W1 - W3
Software Team	Week 3 (4 May)	3 weeks	W3 - W5
Hardware Concept Engineer	Week 3 (4 May)	3 weeks	W3 - W5
Testing & Validation	Week 5 (18 May)	2 weeks	W5 - W6
Documentation & Git Manager	Week 6 (25 May)	2 weeks	W6 - W7

4. Planning Rationale

The ISO 26262 workflow criteria are reflected in the sequential phasing. Before software development starts, safety analysis and architecture must be frozen. Changing ASIL classifications or system boundaries after coding has begun increases the danger of rework. The last part of the project is devoted to testing and validation in order to compare finished hardware and software concepts. In order to prepare the submission package, handle version control, and compile all outputs, the Documentation Manager is activated during the last two

weeks. Throughout the entire seven weeks, the project manager stays involved to manage dependencies and guarantee that milestone deadlines are fulfilled.

5. Ghent Chart

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
								W1 (28-26Apr)	W2 (27Apr-3May)	W3 (4-10 May)	W4 (11-17 May)	W5 (18 - 24 May)	W6 (25- 31 May)	W7 (1 - 7 June)	
SL No	Role	Key Responsibility	No of Member	Job Description	Team Member Names & u-id	Start (Week (20th Apr)	Duration (Week)	1	2	3	4	5	6	7	
1	Project Manager	Planning, Coordination and Final Submission	1	Oversees the entire project, assigns tasks, tracks progress, ensures deadlines are met, integrates all documents, and manages final report and presentation submission.	Rafsun (u41213)	1	7								
2	Safety Analysis Team (TUV Safety Authority)	Hazard analysis, ASIL determination	2	Identifies hazards, evaluates risk (Severity, Exposure, Controllability), determines ASIL level, and defines safety goals based on functional safety standards.	Shakthi Prasad Gangi Reddy(u39436), Nimal Pali (u39349)	1	3								
3	System Architecture Team	System design, block diagram	2	Designs overall ECU architecture including sensors, inputs/outputs, and data flow. Creates system block diagrams and explains system functionality.	Huy Nguyen (u40313) Rafi Prasetya (u40280)	1	3								
4	Software Team	Arduino coding, Logic implementation	2	Develops software logic for reading sensors, decision-making, controlling outputs, and simulating CAN communication while ensuring safety requirements are implemented.	Atashi (u39941)	3	3								
5	Hardware Concept Engineer	Hardware design concept	1	Defines conceptual hardware components such as sensors, ECU board, and interfaces. Specifies input/output ranges and system connections (no physical build required).	Tanmay Ahar (u39421)	3	3								
6	Testing & Validation	Test Planning, Execution, reporting	2	Creates test cases, performs unit/integration/system testing, validates system behavior, and prepares structured test reports including results and analysis.	Partha Pratim Sutradhar (u39352) Ashrafull Islam (u39428)	5	2								
7	Documentation & Git Manager	Documentation, File Organization	1	Maintains all project documents, ensures proper formatting, manages Git repository structure, version control, and organizes all files for final submission.	Shufl U39431	6	2								